

Senior High School



Lesson Exemplar in General Mathematics

Quarter 4

Unit

10

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Lesson Exemplar for General Mathematics
Quarter 1: Unit 1

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Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR

Learning Area:	GENERAL MATHEMATICS	Grade Level:	11
Semester:	SECOND	Quarter:	FOURTH
		Unit:	10

I. LEARNING GOALS

Content Standard	The learners demonstrate knowledge and understanding of compound interest; simple and general annuities; and business and consumer loans
Performance Standard	By the end of the quarter, the learners are able to create a financial proposal using comparisons of different investment plans (e.g. time deposits, mutual funds, pension plans) to recommend and explain the best financial strategy.
Learning Competencies	1. solve problems involving compound interest, including maturity value, future value, nominal rates, and number of terms; 2. illustrate simple annuity, general annuity, and deferred annuity; 3. solve problems involving annuities; 4. calculate the fair market value of a cash flow stream that includes an annuity; 5. differentiate between business and consumer loans and their characteristics, such as repayment terms, requirements, purpose, and interest rates; 6. solve problems involving business and consumer loans (e.g., amortization, mortgage);
II. REFERENCES and MATERIALS	Textbook and Modules Lim, Y., et al. (2016). Math for Engaged Learning. Quezon City, Philippines: Sibs Publishing House. Oronce, O. A. (2016). General Mathematics. Quezon City, Philippines: Rex Book Store. Department of Education. (2016). General Mathematics: Learner's Material. Pasig City, Philippines: Department of Education.

	Verzosa, Debbie Marie B. Ph.D. (2016) Teaching Guide for Senior High School General Mathematics Core Subject, Diliman Quezon City: Commission on Higher Education. Video Lessons https://www.youtube.com/watch?v=Usids2xsqoU https://www.youtube.com/watch?v=OQ9Mv2jwQWo https://www.youtube.com/watch?v=hsj68ZST63g https://www.youtube.com/watch?v=s4XeOrq53Z0 Application Software: Mentimeter https://www.mentimeter.com/ Kahoot Plickers Materials Board, marker, projector, printed scenario (for activities), formula chart, worksheets, quiz paper	
III. CONTENT	Lesson 10.1: Problem Solving Involving Compound Interest (Recommended Time: 2 hours)	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. define compound interest, maturity value, future value, nominal rates and number of terms; 2. solve problems involving, compound interest, maturity value, future value, nominal rates and number of terms.	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge	
	Activity A.1: Leveling Learner Readiness Instructions: Read and analyze the given problem. You want to buy a new smartphone worth ₱15,000, but you don't have enough cash. A friend offers to lend you ₱15,000. Option 1: 10% simple interest after 2 years Option 2: 10% compound interest after 2 years	Activity A.1: Leveling Learner Readiness This activity reviews simple interest from the previous lesson and introduces compound interest through a real-life situation. You may choose from the strategies below based on what best suits your learners' needs and classroom setting. To facilitate, apply one the provided strategies. Strategy 1: Contextualized Learning

	Compute the total payment for each option using the formulas: Simple Interest: Compound Interest: Guide Questions: 1. Determine which option costs more and why. 2. What does the ₱15,000 stand for? 3. Why do the results differ? 4. If you were the borrower, which option would you choose? Why? 5. How can understanding interest help you make wiser financial choices? 6. What values are important when borrowing or lending money?	1. Begin by asking: • “Who among you dreams of buying a new smartphone?” • “If you don’t have enough money, what are your options?” 2. Present the scenario on the board or with the use of a PowerPoint presentation. 3. Explain that understanding interest is important in making financial decisions. 4. Emphasize that this situation reflects a real-life decision young adults often face. 5. Transition by saying: “Let’s find out how much you’ll really pay under two types of interest.” Strategy 2: Collaborative Learning 1. Divide the class into pairs or groups of 3–4. 2. Provide each group with: -a printed copy of the problem or display it on the screen. - a calculator or allow phone calculator use. 3. Instruct them to: -Compute the total payment for Option 1 (Simple Interest) using $I = P \times r \times t$ -Compute the total payment for Option 2 (Compound Interest) using $A = P(1 + r)^t$ 4. Record both results and discuss which option costs more. 5. Encourage the learners to compare their reasoning, not just their answers. 6. Go around to guide, clarify misconceptions, and encourage participation from all members. Strategy 3. Discussion and Reflection 1. Invite groups to present their answers and reasoning.
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2. Write sample computations on the board for comparison.
3. Lead a discussion using these guide questions.

A.2. Establishing the Purpose of the Lesson

Activity A. 2: Appreciating Lesson Relevance

Instructions:

1. Work with your group to discuss the following questions:
 - Which option resulted in a higher total payment?
 - Why do you think that happened?
 - If you were the borrower, how would this affect your choice?
2. Study the situations below. For each case, decide which type of interest: simple or compound is more beneficial, and explain your reason.

Situation	Which type of interest is better?	Why?
A. You are borrowing money to buy a laptop.		
B. You are depositing money in a savings account.		
C. You are lending money to a friend for one year.		

2. Write your group’s answers clearly in the table.
3. Share your ideas with the class.
4. Be ready to explain your reasoning and relate your answers to real-life financial situations.

Activity A. 2: Appreciating Lesson Relevance

In this activity, learners analyze real-life financial situations to compare simple and compound interest and determine which is more beneficial in different contexts. Through group discussion and reasoning, they evaluate how interest type affects borrowing, saving, and lending decisions. The activity helps learners develop practical financial judgment by connecting mathematical concepts to everyday financial choices.

To facilitate apply any of these strategies.

Strategy 1: Reflective Discussion

1. Begin by recalling the first activity’s results:
 - “In our last activity, you computed the total payment for simple and compound interest. Which one cost more?”
2. Allow 2–3 volunteers to share quick answers.
3. Emphasize that both forms of interest can be helpful and disadvantageous depending on the situation.
4. Say: “Now let’s see which type of interest benefits you more in different real-life cases.”

Strategy 2: Collaborative Learning

1. Keep the same pairs or groups from the first activity.
2. Provide a printed or projected copy of the “Which Interest Works for You?” table.

		3. Let groups discuss and decide which type of interest is advantageous in each situation. 4. Encourage them to justify their choices with the reasoning (not just guesses). 5. Move around the room to assist and ask probing questions: • “Why would simple interest be better for borrowers?” • “When might compound interest be an advantage?” 6. Give groups 3-5 minutes to finish. Strategy 3: Contextualized Learning 1. Explain that these examples mirror real financial decisions they might face in borrowing, saving, or lending money. 2. Reinforce connections to real life: • “When you open a savings account or take a loan, the type of interest determines how much you gain or pay.” 3. Ask a few groups to share their answers and explain how they applied the concepts to real-life examples.
B. Instituting New Knowledge	B.1. Presenting Examples	
	Activity B.1 Exploring Key Concepts Instructions Read the problem carefully and do the indicated tasks. Problem Ana deposits P50,000 in Kabayan Savings Bank at 3% compounded annually. How much is her savings after 5 years? Tasks: 1. Identify: • Principal = ₱50,000	Activity B.1 Exploring Key Concepts In this activity, learners apply the compound interest formula to compute the future value of savings over time. By completing a year-by-year table, they observe how interest is added to both the original deposit and previously earned interest. This activity strengthens computational skills while highlighting the advantage of saving early and letting money grow through compounding. The following are suggested strategies for this activity. Strategy 1: Direct Modelling (I do → We do → You do)

- Annual interest rate $r=3\%=0.03$
- Time $t=5$ years
- Compounding: annually

2. Use the compound-interest formula:
 $F=P(1+r)^t$

3. Substitute the values and compute step-by-step:
 $F=50\,000(1+0.03)^5$
 $F=50000(1.03)^5$
 Calculate 1.03^5 then multiply by 50,000.

4. Complete the table by following step 3.

Year (t)	Principal = P Interest rate = r, compounded annually	Principal = ₱50,000 Initial Rate = 3%, compounded annually
1	$P \times (1 + r) = P(1 + r)$	$P50,000 \times 1.03 = 51,500$
2		
3		
4		
5		

5. Compare your result with classmates. Explain in one sentence how compounding made Ana's savings grow.

Guide Questions:

1. How does Ana's savings change each year, and what pattern do you notice in the amount added annually?
2. Why does the interest earned increase even though the interest rate remains the same each year?

1. Show the problem on the board / projector. Read it aloud.
2. Write the formula $F=P(1+r)^t$ and identify P, r, t. Ask a learner to tell each value.
3. Substitute values on the board and compute $(1.03)^5$ either by calculator or year by year model both methods.
4. Explain rounding rules and show the final answer ₱57,963.70.
5. Give one extra problem for the class to solve together (teacher leads steps).
6. Assign one similar problem for pairs to solve independently.

Strategy 2: Guided Discovery (Socratic prompts + calculation)

1. Ask: "If Ana had simple interest instead, how much would she get after 5 years?" (Let pairs compute.)
2. After answers were given, ask: "Compare that with compound outcome. What changed?"
3. Give guiding questions: "What happens to interest in Year 2 that didn't happen in simple interest?"
4. Let groups re-compute using year-by-year multiplication and report observations.

Year (t)	Principal = P Interest rate = r, compounded annually	Principal = ₱50,000 Initial Rate = 3%, compounded annually
1	$P \times (1 + r) = P(1 + r)$	$P50,000 \times 1.03 = 51,500$
2	$P(1 + r) \times (1 + r)$ $= P(1 + r)^2$	$P51,500 \times 1.03 = 53,045$
3	$P(1 + r)^2 \times (1 + r)$ $= P(1 + r)^3$	$P53,045 \times 1.03$ $= 54,636.35$
4	$P(1 + r)^3 \times (1 + r)$ $= P(1 + r)^4$	$P54,636.35 \times 1.03$ $= 56,275.44$

	3. How would Ana's total savings after 5 years be different if the interest were computed using simple interest instead of compound interest?	5	$P(1+r)^4 \times (1+r)$ $= P(1+r)^5$	$P56,275.44 \times 1.03$ $= 57,963.70$
B.2. Discussing the Concept				
	Activity B.2: Deepening Conceptual Understanding Instructions: The following are important terms to remember. Participate in the activities to better understand each term. Definition of terms: Lender or creditor -person (or institution) who invest the money or makes the funds available Borrower or debtor -person (or institution) who owes the money or avails of the funds from the lender Origin or loan date -date on which money is received by the borrower Repayment date or maturity date -date on which the money borrowed or loan is to be completely repaid Time or term (t) -amount of time in years the money borrowed or invested; length of time between the origin and maturity dates Principal (P) -amount of money borrowed or invested on the origin date Rate I -annual rate, usually in percent, charged by the lender, or rate of increase of the investment Interest (I) -amount paid or earned for the use of money	Activity B.2: Deepening Conceptual Understanding In this part, learners will study the key terms and examples related to compound interest. They will learn essential vocabulary and recognize different formulas related to solving compound interest. The following are suggested activities for this part of the lesson. Strategy 1: Discovery Learning Cooperative Learning Word Hunt Puzzle 1. Group the class with 3-5 members each to encourage collaboration. 2. Post a puzzle on the board (printed/ projected). The puzzle includes the words related to compound interest.		

Simple Interest (I_s) -interest that is computed on the principal and then added to it

Compound Interest (I_c) -interest that is computed on the principal and also on the accumulated past interests

Maturity value or Future value (F) -amount after t years that the lender receives from the borrower on the maturity date.

Note: The term t is expressed in years in the formulas. If the given term is in months, days, or between two dates, you need to do some conversions.

If t is given in months, then $t = \frac{\text{number of months}}{12}$

Example: if the term is 39 months, then

$$t = \frac{39}{12} \text{ years}$$

If t is given in days, then $t = \frac{\text{number of days}}{360}$ or $t = \frac{\text{number of days}}{365}$.

Unless otherwise specified, 360 days is used.

If the dates coincide, we count by months. For example, the time between July 20, 2024 and March 20, 2025 is 8 months. Hence, in this case, $t = \frac{8}{12}$.

Maturity Value, Compounding m times a year

$$F = P \left(1 + \frac{r}{m} \right)^{tm} \quad \text{or} \quad F = P(1 + i)^n$$

Where:

F -maturity (future) value

P -principal

r -rate of interest per year

m -frequency of conversion

T	R	E	V	A	U	B	G	L	O	A	P	O	U	N	D	P
S	K	O	O	L	N	O	C	L	F	E	N	O	M	O	S	E
D	B	O	K	K	S	C	H	J	S	O	M	Y	H	A	Y	P
B	X	B	Z	G	O	L	E	T	I	N	A	I	T	S	E	R
Q	U	A	L	P	I	T	B	Y	M	M	T	E	M	B	E	R
S	E	R	G	R	N	C	O	M	P	O	U	N	D	D	C	B
Q	R	P	T	I	O	R	R	P	L	G	R	R	P	E	R	O
F	I	L	E	N	D	E	R	I	E	M	I	S	A	J	S	A
R	A	O	R	C	N	P	O	A	C	I	T	S	G	A	A	M
O	R	A	M	I	V	A	W	L	T	E	Y	C	O	U	L	R
G	R	N	E	P	I	Y	E	F	T	E	U	Q	R		O	L
E	L	D	N	A	N	M	R	U	I	L	E	A	L	I	M	N
F	I	A	T	L	C	E	A	T	M	A	Z	U	N	A	A	C
E	S	T	F	O	S	N	N	U	E	Q		U	Y	O	A	I
R	O	E	Y	I	N	T	E	R	E	S	T	D	A	N	B	H
C	N	I	T	A	N	D	O	E	O	P	O	L	N	I	P	P
F	T	E	U	B	V	I	W	V	R	E	E	F	S	E	G	I
J	Y	O	U	A	I	T	L	A	H	T	P	R	I	N	C	N
O	A	B	V	S	C	O	U	L	T	D	R	I	O	R	N	E
R	E	Y	S	U	S	T	Y	U	P	H	I	E	K	R	A	J
E	V	V	O	L	E	L	X	E	E	T	Y	T	E	A	S	Z

3. Give 3-5 minutes for students to search and highlight/circle the words.
4. After the hunt, call on groups to share words they found.
5. Each time a word is revealed, define it and connect it to the concept of compound interest.
6. Once all words are discovered, present the formula.
7. Explains how the words fit into the formula.

This helps the learners familiarize themselves with key terms and concepts in an engaging way. It also builds

	t -term/time in years $i = \frac{r}{m}$ -interest rate per period $n = tm$ -total number of conversion periods To find the present value P at Compound Interest $P = \frac{F}{\left(1 + \frac{r}{m}\right)^{tm}} \text{ or } P = \frac{F}{(1+i)^n}$ To find the rate r given F, P, and n, we use $r = m \left[\left(\frac{F}{P} \right)^{\frac{1}{n}} - 1 \right]$ To find the number of periods n, we use the formula $n = \frac{\log \log \left(\frac{F}{P} \right)}{\log \log (1 + i)}$ If the length of the term t that is required, we divide n by m: $t = \frac{n}{m}$	curiosity and reinforces vocabulary understanding before the discussion. Strategy 2: Concept Mapping 1. Present all the key terms on the board (e.g., <i>principal, rate, time, interest, maturity value, compound interest</i>). 2. Ask learners to arrange these terms in a concept map showing their relationships. 3. For example, <i>Principal</i> → <i>earns Interest</i> → <i>grows to Maturity Value</i>. 4. Let groups present their maps and explain how each concept connects to compound interest. 5. This helps visual learners clearly understand how different components are related and reinforces the logic behind the compound interest formula. Strategy 3: Interactive Lecture with Real-life Connection 1. Discuss each definition one by one while showing examples (e.g., bank savings, loans, investments). 2. After each term, ask quick check questions like: 3. “Who is the borrower in this case?” 4. “What happens to the money after 2 years?” 5. Allow students to share short, real-life examples related to the term. 6. Discuss the terms and definitions related to compound interest. This encourages participation and helps learners relate mathematical terms to practical, everyday financial decisions.
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	Example: Mang Juan deposits ₱30,000 in <u>MakaPinoy</u> Bank at 5% compounded annually for 4 years. Find the future value and the compound interest. Given: $P = ₱30,000$ $r = 5\% = 0.05$ $m = 1$ (annually) $t = 4$ years Solution: $F = P \left(1 + \frac{r}{m}\right)^{tm}$ $F = 30,000 \left(1 + \frac{0.05}{1}\right)^{(4)(1)}$ $F = ₱36,465.19$ Compound Interest = $36,465.19 - 30,000$ = ₱6,465.19 The future value is ₱36,465.19 and the compound interest is ₱6,465.19 .	
B.3. Developing Mastery		
	Activity B.3: Practicing Learned Skills Instructions: Solve the following problems by following the given example. - Vincent is saving up to buy a motorcycle worth ₱50,000 in 5 years. He plans to invest her money in an e-wallet Save account that offers 6% interest compounded annually. How much should he deposit today so that he can buy her motorcycle after 5 years? - Francis invested 20,000 in Golden Peso Bank, which offers 5% annual interest compounded annually. He wants his money to grow to 30,000. For how many years should Francis keep his money in the bank to reach 30,000?	Activity B.3: Practicing Learned Skills In this activity, learners will firm-up their learning by answering another set of problems. Strategy 1: Gradual Release (“I do, We do, You do”) - Solve a sample problem on the board. Show a step-by-step process and solve it together. - Let the learners practice individually. - Check and give feedback. - Ask: “Why do you think compound interest helps money grow faster than simple interest?” - Class discusses briefly, guided by the formula and word connections This develops the learner’s problem-solving skills through gradual release. This approach ensures that learners gain

	3. Racquel invested 35,000 in two different banks for 4 years, compounded annually: *Golden Peso Bank: After 4 years, her money grew to 45,000. *Mabuhay Savings Bank: After 4 years, her money grew to 47,000. Which bank offered a higher annual interest rate?	confidence and mastery as they move from guided to independent practice. Strategy 2: Pair-and Explain 1. Pair up students and assign each pair one problem from the activity. 2. Each pair solves the problem and explains their solution process to their partner. 3. Encourage them to justify each computation and check if their results match. 4. Select 2–3 pairs to present their solutions to the class for discussion. This promotes peer learning, strengthens reasoning skills, and helps students articulate the process of solving compound interest problems. Strategy 3: Problem Rotation (Math Carousel) 1. Post 3 different compound interest problems around the classroom (each with varying levels of difficulty). 2. Divide the class into small groups and assign each group to a problem station. Give 3–5 minutes per station, then have groups rotate to the next one. 3. After all rotations, review the correct solutions as a class and discuss strategies used. This activity encourages collaboration, application of learned concepts to multiple contexts, and active engagement while reinforcing mastery.
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Making Real-World Connections Instructions:	Activity C.1: Making Real-World Connections In this activity, learners apply their understanding of compound interest to solve authentic, real-life financial problems drawn from everyday Filipino contexts. Working

	Pick one problem (written on the strip of paper) and solve it by group. Discuss your answer and solution with the group and present your output. La Union Cooperative Loan A fisherman in La Union borrowed ₱50,000 from La Union Cooperative at 6% annual interest, compounded semi-annually. How much will he owe after 4? Student Savings for Tuition Marilou wants to have ₱80,000 saved for her college tuition after 5 years. She plans to deposit her money in Golden Peso Bank, which offers 8% compounded annually. How much should she deposit today? Teacher's Pag-IBIG MP2 Investment Mr. Obra, a teacher, invested ₱100,000 in the Pag-IBIG MP2 program, which gives 7% compounded annually. He wants his money to grow to ₱150,000. How many years will it take for his investment to reach ₱150,000? Family Business Fund The Dela Cruz family invested ₱200,000 in a Time Deposit Account for their family business expansion. After 6 years, their savings grew to ₱320,000, compounded annually. What annual interest rate did the bank offer? Overseas Worker's Remittance An OFW in Dubai sends ₱100,000 to his family in the Philippines and advises them to deposit it in Golden Peso Bank with 10% interest compounded annually. After 3 years, the money will be used to renovate their house. How much will the family have after 3 years?	collaboratively, they analyze a chosen situation, compute the required values, and explain their solutions. The activity strengthens problem-solving skills while helping learners see how interest concepts are used in personal finance, investments, and savings decisions. Strategy 1: Collaborative Problem Solving (integrating localized problems) 1. Group the class into 5 groups. Let each group pick one localized scenario problem. 2. Let the group solve the assigned problem. 3. Let the groups appreciate the importance of knowing the concept of compound interest in making a decision. Example: Problem 1: After computing how much the fisherman owes, should he continue borrowing from the cooperative or find another means? Problem 2: After finding how much Marilou deposits now, should she save a lump sum or save smaller amounts monthly? Problem 3: After solving how long it takes to reach ₱150,000, should Mr. Obra stick with MP2 or look for faster investment options? Problem 4: After finding the rate offered, should the Dela Cruz family keep their money in this bank or transfer to another investment that offers higher rates? Problem 5: After computing the future value, is the money enough for the renovation? If not, should the family borrow, save more, or delay the renovation? 4. Let the group present their work written in manila paper or digital slides.
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		5. Then let the class discusses and validates the solution This activity helps reinforce understanding, correct misconceptions, and build confidence as the learners apply what they have learned to more complex situations. Strategy 2: Financial Simulation 1. Group the class into 5 groups. Let each group pick one localized scenario problem. 2. Let the group show the process of solving compound interest and how the interest works through role-playing. 3. Let the group assign roles such as borrower, lender, banker, or investor. Have them act out negotiations or decisions (e.g., choosing between two investment options). 4. Discuss outcomes and identify which options were most beneficial and why. Engages learners through experiential learning, helping them internalize the impact of compound interest on personal and financial decisions.
C.2. Making Generalization		
	Activity C.2: Wrapping up the Lesson Instructions: Answer the question on your notebook and share your answers in class. 1. From all the examples and activities, what can we say about compound interest compared to simple interest? How does compound interest differ from simple interest in terms of how money grows over time? 2. Why is understanding compound interest important when making financial decisions such as saving or borrowing money?	Activity C.2: Wrapping up the Lesson This activity allows learners to reflect on and synthesize their understanding of simple and compound interest based on the lesson and activities completed. By answering guide questions and sharing insights, learners deepen their conceptual understanding and connect mathematical ideas to real-life financial decisions. The activity reinforces the importance of compound interest in managing savings, loans, and long-term financial goals. Strategy 1: Class discussion and synthesis.

		1. Ask guiding questions. 2. List learners' responses. 3. Summarize into a class generalization. 4. Write an agreed statement on the board. This activity helps learners reflect on and consolidate their understanding of compound interest. Through this sharing, learners will connect the lesson to real-life financial situations and deepen their understanding of the concept. Strategy 2: Exit Ticket Reflection 1. Give each learner a small paper or "exit ticket." 2. Ask them to write one thing they learned, one thing they found interesting, and one real-life situation where they can apply compound interest. 3. Collect the papers and read a few responses aloud to summarize class insights. This strategy encourages individual reflection and ensures every student contributes to the generalization process before leaving the class. Strategy 3: One-Sentence Summary 1. Ask learners to summarize what they learned about compound interest in one clear sentence. 2. Allow volunteers to share their sentences aloud. 3. Write a few strong answers on the board, then combine them into a final class generalization. Develops concise communication skills and ensures that learners can express the essence of the lesson in their own words.
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C.3. Evaluating Learning		
	Activity C.3: Assessing Learning Outcomes Instructions: Solve the following problems. Show a complete solution. - Jessa bought a new cellphone worth ₱15,000 using a loan from Golden Peso Bank. The bank charges 10% interest compounded annually for 2 years. How much will she pay after 2 years? - Joy, an OFW, plans to remit ₱150,000 to the Philippines for her family's future needs. She has two investment options for 5 years: Option A (Bank Time Deposit): 5% annual compounding Option B (Barangay Cooperative): 2.5% every 6 months (semi-annual compounding) Which option will yield greater interest after 5 years?	Activity C.3: Assessing Learning Outcomes This activity must be completed individually by each learner. Its completion is essential for assessing their level of understanding of the lesson. Review their work promptly so you can provide timely and appropriate intervention for those who require additional support or remediation.
	C.4. Additional Activities	
	Activity C.4: Extending and Reinforcing Learning Remediation: Solve the following problems. Show a complete solution. - An OFW sent ₱50,000 for his family to invest at 12% compounded annually. After 3 years, how much will the money be worth? - Number of Terms A family deposited ₱30,000 in Golden Peso Bank with an annual interest rate of 5% compounded annually. They want the money to double. How many years will it take for their savings to reach ₱60,000? Enhancement: Solve the following problems. Show a complete solution. - Laptop Savings Plan Ella wants to buy a laptop for college. She decided to save ₱12,000 in Golden Peso Bank which offers 6%	Activity C.4: Extending and Reinforcing Learning This activity provides differentiated tasks to support and extend learners' understanding of compound interest. The remediation problems reinforce foundational skills through guided practice, while the enhancement tasks challenge learners to apply the concept to more complex and varied real-life financial situations. Through these exercises, learners strengthen accuracy, confidence, and practical application in making sound financial decisions. This additional exercise may be given as an assignment. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning.

	interest compounded annually. If she leaves the money for 3 years, how much will she have by the time she buys the laptop? 2. Tricycle Loan Mang Tonyo borrowed ₱80,000 from Golden Peso Bank to buy a tricycle for his livelihood. The bank charges 9% interest compounded annually for 4 years. How much will Mang Tonyo need to pay after 4 years? 3. Fiesta Fund A barangay youth group saved ₱20,000 for their annual fiesta. They deposited it in Golden Peso Bank at 5% compounded annually. If they plan to use it after 2 years, how much will their fiesta fund grow?	
III. CONTENT	Lesson 10.2: Simple and General Annuities (Recommended Time: 2 hours)	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. illustrate simple annuity, general annuity, and deferred annuity; 2. solve problems involving annuities; 3. calculate the fair market value of a cash flow stream that includes an annuity.	
V. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Activating Prior Knowledge	
	Activity A.1: Leveling Learner Readiness Instructions: 1. In pair, compute the final amount you would have in exactly 1 year for each option. 2. Discuss which option is better. Explain why. “Imagine you earned ₱15,000 from your summer job. You want to save this money for a trip next year. You only have 1 year until the trip, so you need to choose the best savings account today.” You approach two different financial institutions with your ₱15,000 lump sum.	Activity A.1: Leveling Learner Readiness This recalls and relates the concepts of interest calculation and periodic payments to prepare for the study of annuities. Strategy: Student-Centered Inquiry Learners aren't just told to calculate interest; they must solve a problem ("choose the best savings account") which drives their learning. 1. Present the problem. 2. Let the learners solve in pairs. 3. Call volunteers to share.

	Option 1. The Old Bank offers 10% annual interest, paid only on the original ₱15,000. Option 2. The Modern Credit Union offers 9.5% annual interest, compounded monthly. Guide Questions: 1. If you were to save your money, which option would you choose? Why? 2. What are the practical benefits of knowing how financial interest works for improving your personal finance choices?	Summarize by highlighting difference between simple and compound interest.
	A.2. Establishing the Purpose of the Lesson	
B. Instituting New Knowledge	Activity A. 2: Appreciating Lesson Relevance Instructions: Reflect on the questions and share your insights. 1. How do people prepare for their retirement? 2. What is the importance of one's knowledge on savings and investment as a preparation for their retirement?	Activity A. 2: Appreciating Lesson Relevance This activity encourages learners to reflect on the real-life relevance of savings and investment concepts, particularly in preparing for retirement. By sharing personal insights and ideas, learners connect mathematical understanding to long-term financial planning and responsible decision-making. Let the students explore their beliefs on the importance of retirement preparation. Use scenarios to aid students. Pose guide questions if necessary.
	B.1. Presenting Examples	
	Activity B.1 Exploring Key Concepts Option1: "The Vacation Fund" Identify the payment interval and compounding period and compare your answer to other learners.	Activity B.1 Exploring Key Concepts This activity helps learners explore key annuity concepts by analyzing realistic financial scenarios related to savings and retirement planning. Through comparison, discussion, and reflection, learners distinguish between different types

	Scenario 1. Sarah wants to save up for a dream vacation. She plans to deposit ₱50,000 at the end of every year for 6 years into a special savings account. The account earns an annual interest rate of 8%, compounded annually. (Simple Annuity) Scenario 2. Sarah wants to save up for a dream vacation. She plans to deposit ₱50,000 at the end of every year for 6 years into a special savings account. The account earns an annual interest rate of 8%, but the interest is compounded quarterly. (General Annuity) Processing Question 1. Based on the given examples, distinguish between simple annuity and general annuity. 2. What are the key differences between a simple annuity and compound annuity? Option 2: “The Retirement Fund” Sarah, now a 35-year-old marketing profession in La Union, wants a reliable source of income to supplement his PAG-IBIG savings when she retires. She was offered the following options: 1. She will make a one-time lump sum payment of P500,000 (single premium deferred annuity) to an insurance company and will receive P10,000 every month for 10 years at the age of 65 onwards (deferred annuity). 2. She will make contributions over 30 years (Age 35 to 35), but with no fixed schedule or amount (flexible premium deferred annuity). The insurance company will compute the total accumulated value; from which she will choose to	of annuities based on payment intervals, compounding periods, and contribution schemes. The activity deepens conceptual understanding while encouraging learners to connect mathematical ideas to long-term financial goals, risk preferences, and personal values in decision-making. Option 1: Collaborative Learning 1. Let the students identify the payment interval and compounding period. 2. Ask the students to compare the findings. Scenario 1. Since the payment interval (annual) matches the compounding period (annually), this is a Simple Annuity. Scenario 2. Since the payment interval (annual) is different from the compounding period (quarterly), this is a General Annuity. Use the situations on the left to elicit definitions from the students. Option 2: Experiential Learning This is an interactive, long-term financial simulation activity that helps students understand the power compound interest, the effects of market fluctuations, the importance of saving early, and the basics of investment diversification. Use the situations on the left to elicit definitions from the students. Based on the given situation on the retirement fund, draw out the terms and let the students describe/define the terms. 1. Deferred annuity
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	receive the value as an income stream every month for 15 years or for the rest of her life at the age of 65 onwards (deferred annuity). Process Questions: - Based from the given situations, describe: - deferred annuity - single premium deferred annuity - flexible premium deferred annuity - When is Option 1 applicable? Option 2? Reflective Question: Considering Sarah's financial goal of supplementing her retirement income, which of the two annuity options—the predictable, fixed payout of Option 1 (Single Premium) or the variable, potentially larger payout of Option 2 (Flexible Premium)—better reflects your personal financial values regarding risk, flexibility, and certainty, and why would you advise Sarah to choose that option today?	- Single premium deferred annuity - Flexible premium deferred annuity Let the students know the applicability of each option. - What is the concept of deferred annuity? - When is option 1 most applicable? Option 2? - Let the students reflect on the given situation by answering the given question.
B.2. Discussing the Concepts		
	Activity B.2: Deepening Conceptual Understanding Game Mechanics: - Learners are grouped (2-4 members). - Each group will be given the set of index cards that contains a brief real-life scenario involving periodic payments and interest. - Each group will be given 5 minutes to sort the given scenario into simple and general annuity. - The first group to do the activity correctly will be the winner.	Activity B.2: Deepening Conceptual Understanding Strategy 1: Card Sort Game This is a highly effective, low-prep activity for concept identification. Game Mechanics: - Preparation: Create a set of index cards, each containing a brief real-life scenario involving periodic payments and interest. - Card Examples:

After the sort, the class reviews each card, and students must verbally identify the Payment Interval and the Compounding Period for verification. This reinforces the core definition.

Given the information above, let us now define:

- Annuity – a sequence of payments made at equal (fixed intervals or periods of time. (Rental payments, Monthly pensions, monthly payment of car loan, educational plan, etc.)

Annuities may be classified in different ways, as follows.

Annuities		
According to payment interval and interest period	Simple Annuity - an annuity where the payment intervals is the same as the interest period	General Annuity - an annuity where the payment intervals is not the same as the interest period
According to time of payment	Ordinary Annuity (or Annuity Immediate) - a type of annuity in which the payments are made at the end of each payment interval	Contingent Annuity - an annuity in which the payments extend over an indefinite (or indeterminate) length of time
According to duration	Annuity Certain - an annuity in which payments begin and end at definite times	Contingent Annuity - an annuity in which the payments extend over an indefinite (or indeterminate) length of time

In the activity, the payment is made at the end of each period. Such annuity is called ordinary or simple annuity.

Each payment in an annuity is called the periodic payment (R).

The time between the payment interval and the last payment interval is called the term of annuity (t).

The sum of the future values of all the payments to be made during the entire term of the annuity is the future value or the amount of an annuity (F).

- Simple Annuity Card: "You pay your car loan monthly, and the interest is compounded monthly." (Payment Interval = Compounding Period)
- General Annuity Card: "You pay your mortgage monthly, but the interest is compounded semi-annually." (Payment Interval $\neq$ Compounding Period)
- Another General Annuity Card: "You deposit money annually into an account that compounds interest quarterly."

3. Gameplay: Students (individually or in teams) are given a stack of cards and two designated sorting areas: "Simple Annuity" and "General Annuity."

4. Discussion: After the sort, the class reviews each card, and students must verbally identify the Payment Interval and the Compounding Period for verification. This reinforces the core definition.

Strategy 2: Real-World Problem Solving

1. Present a Realistic Scenario
2. Define the Problem Clearly
3. Identify the different variables
4. Calculate the given problem.

Strategy 3: Mathematical and Procedural Strategies

The teacher can use the following strategies:

Option 1: Jigsaw Method (Formula Breakdown): Divide the formula into parts and assign each part to a student or pair:

1. **Group 1: Identify Variables** (Find R, r, t, i , and n).
2. **Group 2: Calculate the Growth Factor** $((1+i)^n)$.
3. **Group 3: Calculate the Annuity Factor** $\left[\frac{(1+i)^n - 1}{i} \right]$

The sum of the present values of all payments to be made during the entire term of the annuity is called the present value of n annuity (P)

Annuities may be illustrated using a time diagram. The time diagram for an ordinary annuity is given below.

Time Diagram for an n -Payment Ordinary Annuity						
	R	R	R	R	R...	R
0	1	2	3	4	5	n

Simple Annuity- the payment interval is also the same as the interest period.

Example 1. ₱ 50,000 deposited every year for 5 years at 8% per year compounded annually.

Solution: Notice that ₱ 50,000 was deposited every year and it is compounded annually. Since the compounding period is similar to the payment interval, then this is a type of simple annuity.

Payment	₱ 50,000	₱ 50,000	₱ 50,000	₱ 50,000	₱ 50,000	
Period	0	1	2	3	4	5

The future value of all the payments at the end of term ($t=5$)

4. **Group 4: Final Calculation** ($FV=R \times \text{Factor}$).

- *Goal:* Ensure every student understands the meaning and calculation of each part of the formula.

Strategy 4: Real-Life Application

These strategies connect math to the student's personal financial world.

Option 1: Personalization and Scaling:

Change the scenario to something more immediate or personal to the student, such as saving for a new phone, gaming console, or a concert ticket.

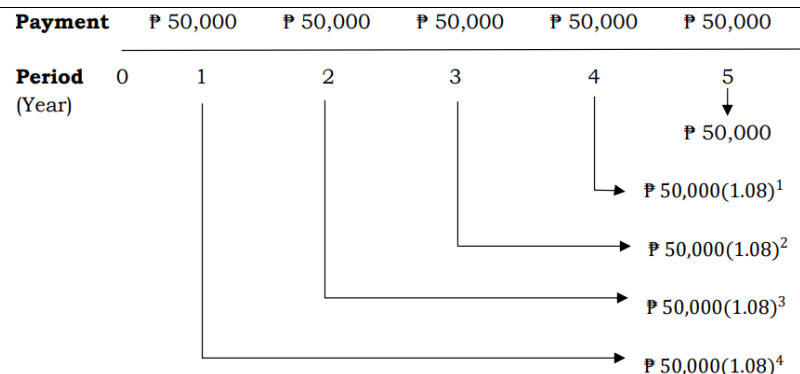
- Ask: "If *you* had a goal of ₱366,796.45 in 6 years, how much would you have to save *monthly* instead of annually?" (This transitions the problem toward finding R for a simple annuity, reinforcing the inverse relationship).

Option 2: Financial Literacy Focus (Interpretation):

After finding the ₱366,796.45 final value, dedicate time to interpreting the result:

- "How much was your own money (₱300,000)? What percentage of the final value is pure interest? ($\approx 18.2\%$)"
- *Goal:* Emphasize the **Time Value of Money** and the importance of investing early and consistently.

(Note: These strategies are also applicable in General Annuity.)



Add all the future values obtained from the previous.

$$\begin{aligned}
 ₱ 50,000.00 &= ₱ 50,000 \\
 ₱ 50,000(1.08)^1 &= ₱ 54,000 \\
 ₱ 50,000(1.08)^2 &= ₱ 58,320 \\
 ₱ 50,000(1.08)^3 &= ₱ 62,985.60 \\
 ₱ 50,000(1.08)^4 &= \underline{₱ 68,024.45} \\
 \text{Total} &= ₱ 293,330.05
 \end{aligned}$$

Example 2: Application of the formula for the Future Value of Simple Annuity
“The Vacation Fund”

Sarah wants to save up for a dream vacation. She plans to deposit ₱50,000 at the end of every year for 6 years into a special savings account. The account earns an annual interest rate of 8%, compounded **annually**. What is the **Future Value (FV)** of Sarah’s annuity? That is, how much money will Sarah have in her account just after making her last ₱50,000 deposit?

Solution

Strategy: Paired Association

Activity: Term-Characteristic Matching

1. A printed material (characteristics of the Single Premium Deferred Annuity (SPDA) and Flexible Premium Deferred Annuity (FPDA) will be distributed to selected members of the group.
2. The students will classify the characteristics as SPDA or FPDA.
3. The responses of the students will be corrected or supplemented.
4. A thorough discussion will be conducted.

Strategy: Explicit Modeling

1. The teacher will present the formulas for the Single Premium Deferred Annuity (SPDA) and Flexible Premium Deferred Annuity (FPDA).
2. Present real-life examples.
3. Illustrate the five-step problem solving strategy.
4. Give a conclusion for each solved problem.

Note:

The teacher may ask the students which among the presented formulas will be applicable to the given problems.

We use the Future Value formula for an Ordinary (Simple) Annuity (payments at the end of the period):

$$FV = R \left[\frac{(1+i)^n - 1}{i} \right]$$

Step 1: Identify the Variables

- **Regular Payment (R):** ₱50,000
- **Annual Rate (r):** 8% or 0.08
- **Compounding Frequency (m):** 1 (annually)
- **Time in Years (t):** 6 years

Step 2: Calculate the Periodic Rate (i) and Total Number of Periods (n)

- Periodic Interest Rate ($i = \frac{r}{m}$):

$$i = \frac{0.08}{1} = 0.08$$

- Total Number of Periods ($n = mt$):

$$n = 1 \times 6 = 6$$

Step 3. Substitute the Values into the Future Value Formula

$$FV = 50,000 \left[\frac{(1 + 0.08)^6 - 1}{0.08} \right]$$

Step 4: Calculate the Future Value Factor

- $(1.08)^6 \approx 1.586874$

$$\text{Future Value Factor} = \frac{1.586874 - 1}{0.08} = \frac{0.586874}{0.08} \approx 7.335925$$

Step 5: Calculate the Final Future Value (FV)

$$FV = 50,000 \times 7.335925$$

$$FV \approx \text{₱}366,796.25$$

Step 6: Answer

The Future Value of Sarah's simple annuity after 6 years is ₱366,796.25.

(Note: She deposited a total of $6 \times \text{₱}300,000$, meaning the interest earned is ₱66,796.25)

General Annuity- refers to an annuity where the length of the payment interval is not the same as the length of the interest compounding period.

Examples of General annuity:

1. Monthly installment of a car, lot or house with an interest rate that is compounded annually.
2. Paying a debt semi-annually when the interest is compounded monthly.

Example 3. Sarah wants to save up for a dream vacation. She plans to deposit **₱50,000** at the **end of every year** for 6 years into a special savings account. The account earns an annual interest rate of 8%, but the interest is compounded **quarterly**. What is the **Future Value** (FV) of Sarah's annuity? How much money will Sarah have in her account just after making her last ₱50,000 deposit?

Key Identification

- **Payment Interval:** Annually (Pmt=1 per year)
- **Compounding Period:** Quarterly (Cmp=4 times per year)

Since Payment Interval $\neq$ Compounding Period, this is a **General Annuity**.

Strategy: Five-Step Problem Solving

1. Identify the required.
2. Identify the given.
3. Give the applicable formula.
4. Solve.
5. Interpret the answers.

Solution: Converting to an Equivalent Simple Annuity

To solve a General Annuity, we must find an **equivalent interest rate** (i_p) that matches the payment period. This involves two steps:

Step 1: Find the Equivalent Rate per Payment Interval (i_p)

We use the formula for the equivalent rate, i_p , which converts the periodic rate of compounding to a rate equivalent to the payment interval.

Given Variables:

- $R = \text{P}50,000$ (Regular Payment)
- $r = 0.08$ (Nominal Annual Rate)
- $m = 4$ (Compounded Quarterly)
- $k = 1$ (Payment Frequency - Annually)
- $t = 6$ years

Formula for Equivalent Rate (i_p):

$$(1 + i_p)^k = \left(1 + \frac{r}{m}\right)^m$$

since $k = 1$ (annual payment), we simplify:

$$1 + i_p = \left(1 + \frac{0.08}{4}\right)^4$$

Calculation:

$$1 + i_p = (1 + 0.02)^4$$

$$1 + i_p = (1.02)^4$$

$$1 + i_p \approx 1.082432$$

$$i_p \approx 1.082432 - 1$$

$$i_p \approx 0.082432 \text{ (This is the effective annual rate)}$$

Step 2. Calculate the Future Value (FV) using the Simple Annuity Formula

Now we treat the problem as a Simple Annuity using the new effective rate (i_p) over the total number of payments (n).

- **Payment (R):** ₱50,000
- **Rate (i):** $i_p \approx 0.082432$
- **Total Periods (n):** 6 (6 annual payments)

Future Value Formula:

$$FV = R \left[\frac{(1 + i_p)^n - 1}{i_p} \right]$$

Substitution and final Calculation:

$$FV = 50000 \left[\frac{(1 + 0.082432)^6 - 1}{0.082432} \right]$$

$$FV = 50000 \left[\frac{(1.082432)^6 - 1}{0.082432} \right]$$

$$FV = 50000 \left[\frac{(1.609460) - 1}{0.082432} \right]$$

$$FV = 50000 \left[\frac{(0.609460)}{0.082432} \right]$$

$$FV = 50000[7.39343]$$

$$FV \approx \text{P}369,671.50$$

Answer: The Future Value of Sarah's general annuity after 6 years is **₱369,671.50**.

(Note: Compared to the original simple annuity (FV of ₱366,796.30 at 8% compounded annually), compounding more frequently (quarterly) with the same nominal rate results in a higher final balance, as expected.)

A **deferred annuity** is a financial contract with an insurance company that allows you to accumulate funds and defer receiving income payment until a future date, typically in retirement.

Type of Deferred Annuity by How the Premium is Paid

1. **Single Premium Deferred Annuity (SPDA)** is a one-time lump sum payment at the beginning of the contract and contributions end immediately after the initial payment. The entire principal starts earning interest and compounding immediately.
2. **Flexible Premium Deferred Annuity (FPDA)** involves multiple payments over time wherein contributions can be made at any time, any amount, as desired. The contributions can be made for many years during the accumulation phase. Compounding starts smaller and grows as you add more to the principal.

The deferred annuity is primarily characterized by two phases:

1. **Accumulated Phase (Deferral Period):** This is the saving and growth period. This is the time when the investment (PV) is made, and no payments are received. The money grows **tax-deferred**, meaning taxes on the earnings are not paid until taking withdrawals or payments starts.

One can contribute money, either as a single lump sum (**SPDA**) or through a series of payment (**FPDA**). It grows based on a compound interest formula. The future value at the end of this phase (FV) becomes the present value for the next phase.

$$FV = PV(1 + i)^k$$

Where:

FV – Future value

PV – Present value

$i = \frac{\text{Annual interest rate}}{\text{compounding periods}}$ ($i =$

k – Number of deferral periods

2. **Payout Phase (Annuitization Phase):** This is the income period. This is the time when the annuity payments (PMT) begin and are received for n periods. At a selected date, the annuity begins making guaranteed income payments, which can last for asset period or for the rest of your life. The

withdrawals or payments you receive are subject to ordinary income tax. The accumulated FV from the first phase in the present value for this annuity.

$$PV_{annuity} = PMT \times \left[\frac{1 - (1 + i)^{-n}}{i} \right]$$

$$PMT = PV_{annuity} \times \left[\frac{i}{1 - (1 + i)^{-n}} \right]$$

Where:

PMT – Payout amount

PV_{annuity} – Present value (accumulated value at an age, FV of the accumulated phase)

i – Payout rate

n – Number of payments

Sample Problems:

Juan, age 45, receives a lump sum of P1,000,000. He decides to purchase a Single Premium Deferred Annuity (SPDA) to supplement his retirement income starting at age 65. How much is the annual payout (PMT) if the annual interest rate is at 5% compounded annually (for the accumulation phase) and the assumed annuity payout rate is at 4% compounded annually (for the payout phase). The payout will last from age 65 to age 80. What is the annual income payment?

Solution:**Step 1: Identify the required quantity.**

- The annual payout (PMT)

Step 2: Identify the given values.

$$PV = P1,000,000$$

Deferral period, k =
20 (from age 45 to age 65)

Annual interest rate, i = 5% or 0.05

Number of payments, n =
15 (from age 65 to age 80)

Annual payout rate, i = 4% or 0.04

Step 3: Formula

Future value, FV (total fund at the age 65)

$$FV = PV(1 + i)^k$$

Payout amount, PMT

$$PMT = PV_{annuity} \times \left[\frac{i}{1 - (1 + i)^{-n}} \right]$$

Step 4: Solution

To compute the annual payout, the future value of the single premium is first calculated.

Future value, FV (total fund at the age 65)

$$FV = PV(1 + i)^k$$

$$FV = 1,000,000(1 + 0.05)^{20}$$

$$FV = P2,653,297.71$$

Payout amount, PMT

$$PMT = PV_{annuity} \times \left[\frac{i}{1 - (1 + i)^{-n}} \right]$$

$$PMT = 2,653,297.71 \times \left[\frac{0.04}{1 - (1 + 0.04)^{-15}} \right]$$

$$PMT = P238,655.72$$

Step 5: Answer

Juan's single premium of P1,000,000 at age 45 would provide him with an annual retirement income of approximately P238,655.72 starting age 65 for 15 years.

1. Juana, 35 years old, decides to purchase a Flexible Premium Deferred Annuity (FPDA) she expects to start receiving payments at age 45. She plans to make varying contributions during the 10-year accumulation period. The table below shows her flexible premium schedule.

Year (k)	Premium Amount (PV)
1	P50,000
2	P60,000
3	P75,000
4-10	P0

The annual interest rate is at 5% compounded annually. The annuitization period is 5 years with an assumed payout rate at 4% compounded annually. What will be the accumulated value and the annual income payment?

Step 1: Identify the required quantity.

- The accumulated value (FV)
- The annual income payment of Juana at age 45

Step 2: Identify the given.

Annual interest rate, $i = 5\%$ or 0.05

Number of payments, $n = 5$

Step 3: Formula

Future value, FV

$$FV = PV(1 + i)^{n-k}$$

Payout amount, PMT

$$PMT = PV_{annuity} \times \left[\frac{i}{1 - (1 + i)^{-n}} \right]$$

Step 4: Solution

To compute the estimated accumulated value (FV), the sum of the future values of the three premiums will be computed.

Year	Premium	Calculation	Future Value
1	P50,000	$50,000(1.05)^9$	P77,566.23
2	P60,000	$60,000(1.05)^8$	P88,674.65
3	P75,000	$75,000(1.05)^7$	P105,627.09
4-10	P0		P0
Total	P185,000		P271,867.97

	The annual income payment is computed as: $PMT = PV_{annuity} \times \left[\frac{i}{1 - (1 + i)^{-n}} \right]$ $PMT = 271,867.97 \times \left[\frac{0.04}{1 - (1 + 0.04)^{-5}} \right]$ $PMT = P61,068.75$ Step 5: Answer Juana's estimated accumulated fund value is P271,867.97. She will receive an annual income payment of approximately P61,068.75 for 5 years, starting at age 45.	
B.3. Developing Mastery		
	Activity B.3: Practicing Learned Skills Instructions: Identify the type of annuity and solve the problems below: 1. Monthly payments of ₱ 2,000 for 5 years with interest rate of 12% compounded annually 2. Yearly payment of ₱15,000 for 10 years with interest rate of 8% compounded annually. 3. Mr. Miguel Dimaano, a 50-year old Filipino professional, purchases an annuity from an insurance company to supplement his retirement income. He will contribute a lump sum of P1,500,000. The annual	Activity B.3: Practicing Learned Skills In this activity, learners apply their knowledge of annuities by identifying the type of annuity in each scenario and solving the corresponding financial problem. Students will compute the future or present value of annuities using the appropriate formulas, reinforcing their understanding of payment schedules, interest rates, and annuitization periods. This practice develops both analytical and computational skills in real-life financial contexts. Answer key:

interest rate and the annual payout rate are 5% and 4%, respectively. The annuitization period is 10 years.

4. Ms. Susan Santos, a 40-year-old marketing executive, purchases insurance from an insurance provider. She decides to make varying contributions during the 15-year accumulation phase.

<i>Year (k)</i>	<i>Premium Amount (PV)</i>
1	P40,000
2	P50,000
3	P50,000
4	P60,000
5-15	P0

The guaranteed annual interest rate is at 4% compounded annually and the assumed payout rate is at 3.5% compounded annually. The annuitization period is 10 years.

- What will be the accumulated value at the end of the 15-year accumulation period?
- How much is the annual income payment during the 10-year annuitization period?

Reflection Questions:

- How did you classify the payments in the given situation above change the way you approached the problem?

1. General Annuity

The total amount available after 5 years is approximately **₱156,161.00**

2. Simple Annuity

The total amount available after 10 years is approximately **₱217,298.44**

3. Deferred Annuity

a. FV = P3,118,392.26

b. PMT = P384,472.23

4. Deferred Annuity

a. FV = P324,203.76

b. PMT = P38,981.82

Strategy: Think-Pair-Share

Phase 1: Think

The students will read the problem and will work out the solution on their paper. The teacher monitors ensuring students are individually engaged.

Phase 2: Pair

	2. How can understanding annuity types help you make wiser decisions about your future investments or loans?	The students turn to an assigned partner and compare their solutions. They will discuss the process and reconcile any differences in their methods. The teacher still monitors and identifies common errors and misconceptions. Phase 3: Share A few student pairs will share and discuss their solutions and answer with the entire class. The teacher facilitates and supplements the discussion.
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Making Real-World Connections Mechanics: A. Group Work (3-5 members): Each group is assigned one localized scenario. B. Problem Analysis and computation: Let the learners solve the assigned problem applying the concept of annuity. C. Decision-Making: Presentation of output applying the assigned strategy for each scenario. Solve the given problem Scenario 1: The Retirement Growth Plan The Situation: You, at age 25, decide to use the MP2 program to save for a future down payment on a larger house. You plan to make regular, voluntary deposits. The Details (Simplified):	Activity C.1: Making Real-World Connections In this group activity, learners apply annuity concepts to solve real-life financial scenarios related to savings and housing loans. Working collaboratively, they analyze the problem, perform computations, and explain their reasoning using the correct annuity type. Finally, each group presents their solution and decision-making strategy to the class, reinforcing practical financial literacy. Answer key: Scenario 1: “The Retirement Growth Plan” 1. The payment interval (annual) matches the interest compounding period (annual). 2. $FV = ₱141,607.72$ Scenario 2: “The First Home Loan” 1. The payment interval (monthly) does not match the compounding period (quarterly). 2. Monthly interest rate: 0.5370% per month 3. $R \approx ₱18,892.45$ Scenario 3 PV remaining $\approx ₱2,128,950.00$

	• Regular Deposit (R): ₱24,000 at the end of every year. • Term (t): 5 years (The minimum holding period for MP2). • Interest Rate (r): Assume a fixed average dividend rate of 7.5% per year. • Compounding/Crediting: Annually (simplified, as dividends are credited annually). The Task (Simple Annuity Mastery): 1. Why is this a simple annuity? 2. Determine the total amount available after 5 years, including earned dividends. Scenario 2: The First Home Loan The Situation: You are applying for a Pag-IBIG Housing Loan to purchase a condo unit. The loan amount you need is determined by the maximum monthly payment you can afford. The Details: • Loan Amount/Present Value (PV): ₱3,000,000 • Loan Term (t): 30 years. • Payment Frequency (k): Monthly (12 payments per year). • Interest Rate (r): 6.5% per year. • Compounding Frequency (m): Quarterly (4 times per year).	Scenario 1: The Retirement Growth Plan 1. Strategy: Contextualized/Situational Learning • Focus: Students work with problems embedded in realistic financial scenarios (MP2 savings, Pag-IBIG housing loans). • Goal: This makes the abstract mathematical concepts relevant and tangible, answering the perpetual "When will I ever use this?" question. It directly connects mathematics to financial literacy and real-life decision-making. 2. Mastery Learning: • Focus: The problems are structured to ensure deep understanding of specific concepts ("Simple Annuity Mastery," "General Annuity Mastery"). • Goal: Students must demonstrate a high level of competence (mastery) in one concept before moving to the next. The progression from Scenario 1 (Simple Annuity) to Scenario 2 (General Annuity) and finally to Scenario 3 (Integration/FMV) is a perfect example of scaffolding towards mastery. A. Group Work (3-5 members): Each group is assigned one localized scenario. B. Problem Analysis and computation: Let the learners solve the assigned problem applying the concept of annuity. C. Decision-Making: Presentation of output applying the assigned strategy for each scenario. Scenario 1(Task):
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The Task (General Annuity Mastery):

1. Why is this a General Annuity?
2. Find the monthly interest rate that is financially equivalent to 6.5% compounded quarterly.
3. What is the Present Value in the given problem? Use the present value formula, rearranged to solve for R.

Scenario 3. Fair Market Value Integration

To meet the third objective (calculating Fair Market Value, FMV), the housing loan scenario can be extended:

After 10 years (120 payments), you receive a bonus and decide to pay off the remaining loan. The bank gives you the option to pay a single lump sum today. Calculate the FMV (Fair Market Value) of the remaining cash flow (the last 20 years, or 240 payments). This FMV is your **remaining principal balance**.

Scenario 4: "OFW Retirement"

Case Prompt: A returning OFW, Manuel, is 40 years old. He has saved P1,000,000. He plans to use this for retirement, which starts at age 65. He wants to receive a monthly income for 15 years, starting at age 65.

Questions for Discussion:

1. Why can't Manuel rely on his Social Security System (SSS) pension?
2. Why is the money deferred?
3. What is the benefit of letting it sit for 25 years?

1. Vocabulary and Classification: Start with a Collaborative Learning on Task 1 ("Why is this a simple annuity?"). Students define "Simple Annuity" and justify the classification based on the criteria ($k=m$).

2. Formula Derivation/Application: Use a Guided Practice approach to show the Future Value formula

$$FV = R \left[\frac{(1+i)^n - 1}{i} \right]$$

explaining each variable ($i = \frac{r}{m}$, $n = mt$).

Scenario 2: The First Home Loan

1. Concept Contrast (Jigsaw Activity): The group will explain the difference between a Simple and a General Annuity (Task 1).

2. Financial Equivalence: Focus on Task 2 using a Step-by-Step Modeling strategy to derive the equivalent periodic rate (i_p) using the equivalence formula:

$$(1 + i_p)^k = \left(1 + \frac{r}{m}\right)^m$$

This is a crucial conversion step in general annuities.

- Problem-Solving for R: Use a Working Backwards strategy for Task 3. Given PV, students must first solve for the equivalent rate (i_p) and then substitute it into the Present Value formula to solve for the regular payment (R).

Scenario 3: The Fair market value **integration**

“What If” Scenarios

Once the base case is solved, explore the decision-making process by adjusting one variable, the deferral period.

1. If Manuel needs to start paying out at age 55 (10 years earlier), what would happen to the annual income?
2. What if the deferral period is increased? Decreased?

Rubrics

Criteria	4 (excellent)	3 (Proficient)	2 (Developing)	1 (Beginning)
Annuity Classification & Justification	Accurately identifies the annuity type and provides a clear, concise, and provides a clear justification based on payment and compounding frequencies	Correctly identifies the annuity type and provides a correct, yet general, justification.	Identifies the annuity type but the justification is weak or contains minor technical errors.	Incorrectly identifies the annuity type or provides no valid justification.
Mathematical Accuracy & Computation	Shows all steps of the Future Value calculation clearly, uses the correct formula and variables, and arrives at the perfectly accurate final amount.	Shows the steps and uses the correct formula, but has one minor computational error leading to a slightly inaccurate final answer.	Uses the correct overall approach but contains significant computational errors or uses an inappropriate formula structure.	Uses an incorrect formula or approach; calculation is largely inaccurate or incomplete.
Problem Analysis & Interpretation	Clearly and insightfully interprets the final answer by relating it back	Correctly states the meaning of the final answer and relates it to the	States the final answer but offers only a minimal or vague	Does not interpret the final answer or misinterpret

1. Real-World Simulation (Role-Playing): Have students take on the roles of the **homeowner** and the **bank loan officer**. The homeowner asks, "How much do I owe today to pay off the rest of the loan?" The loan officer calculates the **FMV** (remaining principal).

2. Abstraction via Timeline: Use a **Visual Timeline** (a powerful tool in finance) to clearly show the 30-year term, the 10 years paid, and the **20 years/240 payments remaining**. The FMV is the Present Value of those *remaining* 240 payments, discounted back to the end of year 10. This links future cash flows to a present value.

Scenario 4:

Strategy: Case Analysis

Activity: OFW Retirement

The group will answer the given problem and present their output in the class.

		to the client's goal. Demonstrate a deep understanding of interest earned vs. principal.	scenario goal, showing a good understanding of the result.	interpretation of its relevance to the financial goal.	the meaning of the Future Value calculation.
	Group Work & Presentation	Presentation is highly engaging, logically structured, and demonstrates cohesive group understanding of the concept. All group members contribute equally and effectively.	Presentation is clear, well-structured, and demonstrates adequate group understanding. Most group members contributed effectively.	Presentation lacks organization or clarity. Evidence of uneven participation among group members.	Presentation is disorganized, confusing, and does not clearly articulate the solution or analysis.
	Total Points: 4 points X 4 criteria = 16+ maximum points Grading Scale: 15-16 points: Outstanding Performance 13-14 points: Strong Performance 10-12 points: Satisfactory Performance Below 10 points: Needs Improvement				
	C.2. Making Generalization				
Activity C.2: Wrapping up the Lesson Activity1: “Match or Mismatch”				Activity C.2: Wrapping up the Lesson This closing activity helps learners consolidate their understanding of annuities by applying key concepts in a quick and interactive way. Through matching scenarios and answering reflective questions, students demonstrate mastery of payment vs. compounding intervals.	

Instruction:

Given four scenarios.

1. Prepare a sheet of paper with two columns: "Payment Interval" and Compounding Interval.
2. Match the payment interval to the compounding interval.
3. Classify if the given scenario is simple or general annuity.

Scenario	Payment Interval	Compounding Interval
A	Monthly Loan Payment	Interest Compounded Monthly
B	Quarterly Deposit	Interest Compounded Annually
C	Annual Premium	Interest Compounded Annually
D	Monthly Loan Payment	Interest Compounded Semi-Annually

Activity 2:

1. Every deferred annuity problem can be broken into two distinct financial math problems. What is the single point in time that acts as the financial bridge, connecting the accumulated stage to the payout stage?
2. At the end of the deferral period, the lump sum saved plays two roles simultaneously. How is the lump sum viewed by the accumulation period,

differentiate between simple and general annuities, and deepen their understanding of deferred annuities and their two-stage structure. This activity also encourages learners to make connections between mathematical procedures and real-life financial decisions.

Strategy 1: Think-Pair-Share:**Activity 1: "Match or Mismatch"**

This activity is a specific structured classroom activity used for review or introducing new material.

1. Ask the students to choose their pair.
2. Prepare a sheet of paper with two columns: "**Payment Interval**" and "**Compounding Period.**"
3. For each one, students must identify the interval for both payment and compounding.
4. After recording the intervals, guide the students to make the final classification.
5. Choose volunteer learners to give the definition of simple and general annuity based on the given activity.

Answers:

	and how is the exact amount viewed by the payout period? 3. If two people invest the exact same amount and want the same payout amount, but one person defers payments 5 years longer, who needs a smaller initial investment today? 4. From the activities and discussions, make a statement that covers one of the concepts or processes.	1. The end of the deferral period 2. Future value; present value of annuity 3. The person with the longer deferral period needs a smaller initial investment. 4. The students will write their answers in their Math journal. Note: The students may brainstorm as they answer the questions.
C.3. Evaluating Learning		
	Activity C.3: Assessing Learning Outcomes A. Please read the sentences carefully and fill in the missing word/s by writing your answer on the line/s provided. 1. A sequence of payments made at equal (fixed) intervals or periods of time is called _____. 2. Each payment in an annuity is called _____. 3. The time between the successive payments dates of an annuity is called the _____. 4. In simple annuity the payment interval is also the same as the _____. 5. In general annuity the interest conversion or compounding period is _____ with the payment interval. B. Solve the following problems. Follow the five-step process.	Activity C.3: Assessing Learning Outcomes This activity evaluates learners' understanding of annuities through both vocabulary and application-based problems. Students demonstrate mastery by completing a fill-in-the-blank review and solving real-life annuity scenarios using the five-step problem-solving process. The activity helps ensure that learners can accurately identify annuity types and compute values for both accumulation and payout stages. Key Answer: A. 1. Annuity 2. Periodic payment or regular payment 3. Payment interval 4. Interest period or compounding period 5. different B. 1. FV= ₱7,052.88

1. Your mom asked your opinion if she will be joining the cooperative in her office. She wants to know the amount of money she will be receiving after 6 months and wants to buy something in December. The cooperative wants her to contribute ₱1,000 per month beginning in June 2020 which will earn 3% compounded monthly. How much will be the future value of your mom's contribution at the end of December 2020
2. Mr. Jose dela Cruz, a 45-year-old OFW, just received a retirement bonus of P1,500,000. He decides to use this lump sum to purchase SPDA to supplement his retirement income starting at age 65. The insurance offers a guaranteed rate of 4.5% per annum, compounded quarterly, during both deferral and payout phases. He wants his annuity to provide him with monthly retirement income for 15 years.
 - a. What is the accumulated value of the single premium?
 - b. What is the guaranteed monthly payout Mr. dela Cruz will receive starting at age 65?
3. Ms. Ana Reyes, a 45-years old overseas Filipino worker (OFW), decides to use a portion of her savings to purchase an annuity from a Philippine insurance company to ensure an additional retirement income starting at age 60. She will contribute an amount of P1,000,000. The annual interest rate is at 4.5% and the payout rate is at 4%. The guaranteed payout period is 10 years.

2. a. $FV = P3,662,997.43$
b. $PMT = P27,711.23$ (monthly)
3. a. $FV = P1,935,282.00$
b. $PMT = P238,604.59$

Note:

The students will answer the problems independently.

	a. What will be the accumulated value of the single premium at the end of the 15-year deferral period? b. What is the amount of the annual income payment Ms. Reyes will receive for 10 years during the payout period?	
C.4. Additional Activities		
	Activity C.4: Extending and Reinforcing Learning Remediation Solve the following problems. Show a complete solution. 1. A man wants to save P10,000 per year for 10 years. If he can earn 10% interest annually on his investments, what will be the accumulated amount at the end of 10 years? 2. A high school teacher, Ms. Cruz, deposits ₱3,000 at the end of every quarter into a retirement fund. The fund earns 4% interest compounded quarterly. How much money will she have after 10 years? 3. A recent graduate wishes to receive P500 at the end of every month for 4 years to cover living expenses while in graduate school. If the money can earn 6% compounded monthly, how much must they deposit today? 4. Mark deposits ₱1,000 at the end of every month for 5 years. The account earns 6% interest compounded annually. What is the final amount in the account? Enhancement Solve the following problems. Show a complete solution. 1. Simple Annuity: Goal Seek (Finding the Rate) Scenario: You inherited ₱100,000 and decide to deposit ₱2,000 at the end of every month into a savings plan. Your goal is to reach a Future Value (FV) of ₱360,000 in exactly	Activity C.4: Extending and Reinforcing Learning This activity provides differentiated practice to reinforce and extend learners' understanding of annuities through real-life financial problems. The remediation set strengthens foundational skills using straightforward annuity computations, while the enhancement set challenges learners with higher-order tasks such as solving for interest rates, adjusting payment schedules, and working with deferred annuities. Students are required to show complete solutions to demonstrate mastery of the concepts and procedures This additional exercise may be given as an assignment. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning. Answer key: Remediation: 1. ₱159,374.24. 2. $FV \approx ₱146,659.11$ 3. $PV \approx ₱21,290.15$ 4. $FV \approx ₱71,670.50$ Enhancement:

	10 years. What annual interest rate (compounded monthly) must your savings plan achieve to meet this goal? 2. General Annuity: The Payment Swap Scenario: A debt requires an annual payment of ₱50,000 for 5 years at 7% interest compounded semi-annually. The borrower wants to change the schedule to quarterly payments. What is the new required quarterly payment (Rnew)? 3. Deferred Annuity: Finding the Initial Investment Scenario: A client wants to receive \$1,200 at the end of every month for 10 years starting 7 years from today. The account earns 6% compounded monthly. How much must be deposited today?	1. The plan must achieve an annual rate of approximately 9.40% (compounded monthly). 2. The new required quarterly payment is approximately ₱12,398.92. 3. The client must invest \$75,640.45 today.
III. CONTENT	Lesson 10.3: Business and Consumer Loans (Recommended Time: 2 hours)	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. illustrate business and consumer loans; 2. differentiate the characteristics between business and consumer loans, such as repayment terms, requirements, purpose, and interest rates; and 3. solve problems involving business and consumer loans (e.g., amortization, mortgage).	
V. PROCEDURES	ACTIVITIES	ANNOTATION
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge	
	Activity A.1: Leveling Learner Readiness Option 1: Quick Poll / Show of Hands Instructions: 1. Raise your hand if you or your family have heard or used the word “loan.” 2. Raise your hand if you know someone who has borrowed money to buy something or start a business.	Activity A.1: Leveling Learner Readiness This readiness activity engages learners through various interactive options to activate prior knowledge about loans and borrowing. Students participate in quick polls, discussions, sorting, word association, or an energizer game to identify and differentiate loan-related situations. The activity prepares learners for the lesson by connecting

	Option 2: Think–Pair–Share Instructions: Think: Imagine you want to buy something expensive (e.g., motorbike, laptop) but don't have enough money. What will you do? Pair: Share your answer with a classmate. Share: Two pairs combine and tell the class what they discussed. Option 3: Sorting Activity (Loan or Not a Loan?) Instructions 1. Read the cards/situations given. 2. Decide whether each one is Loan or Not a Loan. Option 4: Word Association Web 1. On the board, write "Loan." 2. Call out words you think of when you hear "Loan." Option 5: Energizer Game: Loan or No Loan? 1. Listen to each situation. 2. If it involves a loan, shout "LOAN!" or raise a green card/thumbs up. 3. If not, shout "NO LOAN!" or raise a red card/thumbs down.	everyday experiences to the financial concepts they will study. The following suggested activities are designed to help us explore and understand our students' own definitions and perceptions of loans more deeply. You may select the one that best fits your class. Option 1: Questioning / Class Survey 1. Ask two simple yes-or-no questions about loans. 2. Observe how many students raise their hands. 3. Use responses to introduce the relevance of the lesson. Option 2: Collaborative Learning 1. Pose a real-life question to stimulate thinking. 2. Give 1–2 minutes of silent reflection. 3. Allow pairs to exchange ideas. 4. Call on some pairs to share with the class and connect to the topic. Option 3: Experiential / Categorization Strategy 1. Prepare simple scenario cards. <i>Sample situation on the cards:</i> -Juan borrowed ₱50,000 from the bank to start a small computer shop. (LOAN) -Maria used her savings to buy a new washing machine for their house. (NOT LOAN) -Pedro applied for a housing loan to buy a new home for his family. (LOAN) -Ana received a scholarship grant to pay for her tuition fees. (NOT LOAN)
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		<i>-Liza used her credit card to buy groceries for the week. (LOAN)</i> 2. Distribute them to groups or read aloud. 3. Ask groups to classify and justify their answers. 4. Confirm and explain the correct classifications. Brainstorming / Concept Mapping 1. Write the word “Loan” at the center of the board. 2. Record student responses around the central word. 3. Connect and highlight key terms (e.g., money, bank, interest, repayment). Game-Based Learning 1. Prepare 5–6 short situations. <i>(Sample situations are given in option 3)</i> 2. Explain the rules (shout or raise cards). 3. Read each situation aloud with energy and pace. 4. Give immediate feedback after each answer.
A.2. Establishing the Purpose of the Lesson		
	Activity A. 2: Appreciating Lesson Relevance Instructions: Listen to the question and share your ideas: “Why do people or businesses borrow money?” Can you think of a situation when borrowing is helpful and when it is risky?”	Activity A. 2: Appreciating Lesson Relevance This activity helps learners understand the real-life importance of borrowing and lending by reflecting on why individuals and businesses take loans. Through guided discussion, students identify situations where borrowing can be beneficial and when it can become risky. The activity builds relevance by connecting financial concepts to practical decision-making in everyday life. Strategy 1: Reflective Learning Let the learners reflect on and share the importance of the lesson. Encourage them to express their realizations on how understanding the difference between needs and wants can guide responsible borrowing. Highlight that

		prioritizing essential needs such as education, livelihood, or business growth over impulsive desires can lead to wiser financial choices and prevent unnecessary debt. Strategy 2: Math Journaling Students will write their personal reflections, after which the teacher may randomly select and read some entries to highlight diverse perspectives.
B. Instituting New Knowledge	B.1. Presenting Examples	
	Activity B.1 Exploring Key Concepts Instructions: In the previous activity, we identified situations that show whether an example is a loan or not a loan. This time, let us take a closer look and analyze whether each situation involves a business loan or a consumer loan. Identify which is/are business or consumer loan. • <i>Juan borrowed ₱50,000 from the bank to start a small computer shop. → Business Loan</i> • <i>Pedro applied for a housing loan to buy a new home for his family. → Consumer Loan</i> • <i>Liza used her credit card to buy groceries for the week. → Consumer Loan</i>	Activity B.1 Exploring Key Concepts In this activity, learners deepen their understanding of loans by classifying real-life borrowing situations into business loans or consumer loans. By analyzing examples and identifying the purpose of borrowing, students distinguish between loans taken for income-generating ventures and those for personal needs. This activity reinforces the concept of loan types and their relevance to financial decisions. Use the situations on the left to elicit definitions from the students. Select a strategy that best suits your class: Option 1: Think-Pair-Share 1. Ask students to think silently whether it is a business or consumer loan. 2. Let them pair up and share their reasoning with a seatmate. 3. Invite pairs to share their answers with the class to validate ideas. Option 2: Sorting Activity (Card or Board Work) 1. Prepare strips of paper or digital slides showing loan situations.

		2. Call on students to classify each example under Business Loan or Consumer Loan on the board or chart. 3. After sorting, discuss the reasons behind their placements.
B.2. Discussing the Concept		
	Activity B.2: Deepening Conceptual Understanding Based on the examples, let's define: • Business Loan-money lent specifically for a business purpose. It may be used to start a business or to have a business expansion. • Consumer Loan- money lent to an individual for personal or family purpose • Term of Loan-Time to pay the entire loan • Amortization Method- Method of paying a loan (principal and interest) on installment basis, usually of equal amounts at regular intervals • Mortgage- a loan, secured by a collateral, that the borrower is obliged to pay at specified terms • Outstanding balance- Any remaining debt at a specified time Characteristics of Business Loans Purpose: Borrowed mainly to start, maintain, or expand a business (e.g., capital, equipment, raw materials). Repayment Terms: Often flexible but can be long-term (monthly or quarterly payments, depending on business income). Requirements: Usually require business plans, financial records, collateral, and good credit history.	Activity B.2: Deepening Conceptual Understanding This activity deepens learners' understanding of loans by defining key terms, comparing business and consumer loans, and analyzing loan characteristics. Through guided examples and step-by-step problem-solving, students apply formulas for interest, mortgage, amortization, and outstanding balance. The activity strengthens financial literacy and prepares learners for real-life loan decisions by showing how loans are computed and repaid over time. Option 1: T-Chart / Graphic Organizer 1. Draw a T-Chart on the board with two columns: <i>Business Loan</i> and <i>Consumer Loan</i>. 2. Provide lists of characteristics of loans 3. Ask students to place each characteristics under the correct column (business or consumer) 4. Discuss as a class and summarize the differences in a visual format for easy reference. Option 2: Interactive Whiteboard / Digital Sorting 1. Use digital tools (Google Jamboard, Padlet, or a slide) to create draggable cards with loan examples and characteristics. 2. Students drag each card to Business Loan or Consumer Loan columns.

Interest Rates: Can be lower than consumer loans because they are often secured and intended to generate income.

Characteristics of Consumer Loans

Purpose: Borrowed for personal or household use (e.g., buying a house, car, appliances, or paying for education).

Repayment Terms: Typically fixed, with monthly installments over a set period.

Requirements: Generally simpler than business loans; may only require proof of income, employment, and credit history.

Interest Rates: Often higher than business loans, especially if unsecured, since repayment depends on personal income rather than business profit.

Sample Problems

1. Mr. Garcia borrowed P1,000,000 for the expansion of his business. The effective rate of interest is 7%. The loan is to be repaid in full after one year. How much is to be paid after one year?

STEP 1: Identify what is being asked

The value of the loan after 1 year

STEP 2: Identify the Given values

Given: $P=1,000,000$

$j=0.07$

$F=?$

STEP 3: Formula

$$F = P(1 + j)^n$$

STEP 4: Solution

$$\begin{aligned} F &= P(1 + j)^n \\ &= 1,000,000(1 + 0.07) \end{aligned}$$

3. Class reviews placement and discusses differences in real-time.

Option 3. Matching Activity

1. Prepare cards or slips with terms on one set (e.g., Business Loan, Consumer Loan, Purpose, Repayment Terms, Requirements, Interest Rate).
2. Prepare another set with definitions or examples.
3. Students work in pairs or groups to match terms to their correct definitions or examples.
4. Review answers and clarify any misunderstandings.

For Problem solving the following strategies would be suggested:

Strategy 1: Real-World Problem Solving

1. Present a Realistic Scenario
2. Define the Problem Clearly
3. Gather and Analyze Relevant Information
4. Plan a Solution Strategy
5. Execute the Plan
6. Evaluate and Interpret the Solution
7. Reflect on Learning

Strategy 2: Scaffolding

1. Introduce the Problem
2. Activate Prior Knowledge
3. Break the Problem into Smaller Parts
4. Model the First Few Steps
5. Provide Guided Practice
6. Encourage Independent Problem Solving
7. Check and Discuss Solutions
8. Reflect and Summarize

$$= 1,070,000$$

STEP 5: Answer:

An amount of 1,070,000 must be paid after 1 year

2. (Chattel Mortgage) A person borrowed P 1,200,000 for the purchase of a car. If his monthly payment is P31,000 on a 5-year mortgage, find the total amount of interest.

STEP 1: Identify what is being asked

The total amount of interest of a purchased car

STEP 2: Identify the Given values

Given: $P=1,200,000$

Monthly payment=31,000

Term of loan =5 years

Total amount of interest=?

STEP 3: Formula

Total interest = Total amount paid – the amount of the mortgage

Where: Total amount = Amortization x term of loan

STEP 4: Solution

$$\begin{aligned}\text{Total amount} &= (31,000)(12 \text{ months})(5 \text{ yrs}) \\ &= 1,860,000\end{aligned}$$

$$\begin{aligned}\text{Total interest} &= 1,860,000 - 1,200,000 \\ &= 660,000\end{aligned}$$

STEP 5: Answer:

The total amount of interest of a purchased car is 660,000

3. If a house is sold for P3,000,000 and the bank requires 20% down payment, find the amount of the mortgage.

STEP 1: Identify what is being asked
The amount of the mortgage.

STEP 2: Identify the Given values
- Total price of the house: ₱3,000,000
- Down payment: 20% of the total price

STEP 3: Formula
Mortgage Amount = Total Price - Down Payment
Down Payment = Total Price × Down Payment Rate

STEP 4: Solution
Down payment = ₱3,000,000 × 0.20
= ₱600,000
Mortgage amount = ₱3,000,000 - ₱600,000
= ₱2,400,000

STEP 5: Answer:
The mortgage amount is P2,400,000.00

4. A consumer loan worth P80,000 is to be repaid in 12 months at 8% convertible monthly. How much is the monthly payment?

STEP 1: Identify what is being asked
R = The monthly payment of the loan.

STEP 2: Identify the Given values
A = 80,000
 $j = \frac{i^{12}}{12} = \frac{0.08}{12}$
n = 12 (number of payments)

STEP 3: Formula
$$R = \frac{P}{\frac{1 - (1 + j)^{-n}}{j}}$$

STEP 4: Solution

$$R = \frac{\frac{P}{1 - (1+j)^{-n}}}{j} = \frac{80,000}{\frac{1 - \left(1 + \frac{0.08}{12}\right)^{-12}}{\frac{0.08}{12}}} = 6,959.07$$

STEP 5: Answer:

The monthly payment is P6,959.07

5. (Outstanding Balance) Mrs. Se borrowed some money from a bank that offers an interest rate of 12% compounded monthly. His monthly amortization for 5 years is P11,122.22. How much is the outstanding balance after the 12th payment?

STEP 1: Identify what is being asked

The outstanding balance after the 12th payment.

STEP 2: Identify the Given values

$$R = \text{P}11,122.22$$

$$j = \frac{i^{12}}{12} = \frac{0.12}{12} = 0.01$$

$$n = 60 \text{ (number of payments} = 12 \times 5)$$

$$k = 12 \text{ (number of payments paid)}$$

STEP 3: Formula

(Outstanding balance is denoted by B_K)

$$B_K = R \left[\frac{1 - (1+j)^{-(n-k)}}{j} \right]$$

STEP 4: Solution

$$\begin{aligned} B_K &= R \left[\frac{1 - (1+j)^{-(n-k)}}{j} \right] = 11,122.22 \left[\frac{1 - (1.01)^{-(48)}}{0.01} \right] \\ &= 422,354.73 \end{aligned}$$

STEP 5: Answer:

The outstanding balance is P422,354.73

B.3. Developing Mastery

Activity B.3: Practicing Learned Skills

Instructions:

Solve the following:

1. Mr. and Mrs. Banal purchased a house and lot worth P4,000,000. They paid a down payment of P800,000. They plan to amortize the loan of P3,200,000 by paying monthly for 20 years. The interest rate is 12% convertible monthly.
 - (a) How much is the monthly payment?
 - (b) What is the total interest paid?
2. Mrs. Rosal bought a car. After paying the down payment, the amount of the loan is P400,000 with an interest rate of 9% compounded monthly. The term of the loan is 3 years. How much is the monthly payment?
3. Mrs. Tan got a business loan worth P800,000. She promised to pay the loan semi-annually in 5 years. The semi-annual payment is P103,603.66 if money is worth 10% converted semi-annually. How much is the outstanding balance after the third payment?

Activity B.3: Practicing Learned Skills

In this activity, learners apply their knowledge of loans and amortization to solve real-life financial problems involving house, car, and business loans. Students compute monthly or semi-annual payments, total interest, and outstanding balance using appropriate formulas and the five-step problem-solving process. This activity strengthens learners' ability to analyze loan terms and make informed financial decisions.

Strategy: Computer Aided Instruction:

1. Kahoot Technology

Use Kahoot to create and input quiz questions. Students can then respond interactively using their devices, making the activity more engaging and dynamic.

2. Plickers

Use Plickers to assess students' understanding in real-time without the need for individual devices. Insert these questions in advance and have students respond using printed cards, which you can scan with your mobile device.

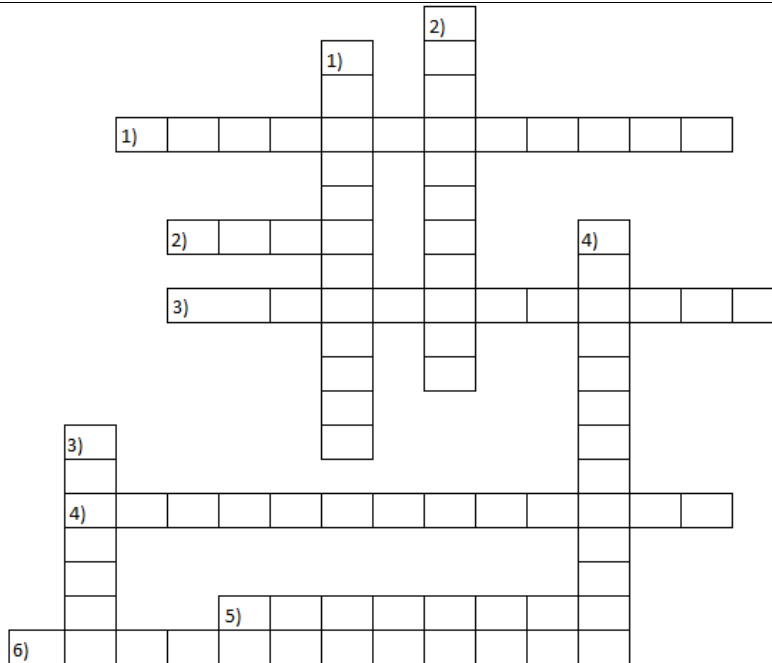
Key Answers:

1. (a) the monthly payment is P35,234.76
(b) the interest amount is P5,256,342.40
2. The regular payment is P12,719.89
3. The outstanding balance after the third payment is P599,489.46

C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Making Real-World Connections Below are four activity options. Choose one activity that matches your interest. Then, find classmates who have chosen the same activity to form your group. Option 1: Role Play 1. Form groups of 5-6 members. 2. Assign roles: one student as the loan officer, one as the client, and others as family/friends. 3. The client presents a need (e.g., buying a house, starting a small business). 4. The loan officer must: • Identify if it is a Business Loan or Consumer Loan. • Explain the characteristics (purpose, repayment terms, requirements, interest rate). • Show a sample computation of a monthly payment. 5. Present your role play to the class in 3–5 minutes. Option 2: Poster / Infographic 1. Work individually or in pairs. 2. Create a poster or infographic comparing Business Loans vs. Consumer Loans. 3. Your work must include: • Purpose • Repayment Terms • Requirements • Interest Rates 4. Add a solved example problem (e.g., mortgage or consumer loan monthly payment). 5. Be creative! Use colors, drawings, or digital design to make your work clear and engaging.	Activity C.1: Making Real-World Connections Strategy: Differentiated Instruction 1. Present multiple activity options (e.g., role play, poster/infographic, song/jingle, storyboard/comic strip) that address the same competencies on business and consumer loans, ensuring tasks cater to varied learning styles and intelligences. 2. Allow students to choose the activity that best matches their strengths and interests, encouraging ownership of learning and intrinsic motivation. 3. Form groups based on chosen activities (5–6 members per group, or pairs for poster-making) to promote collaboration while still respecting individual preferences. 4. Clarify instructions and expected outputs for each option by providing detailed guidelines and examples to avoid confusion and ensure fairness. 5. Share the assessment rubric with students so they understand how their outputs will be evaluated in terms of content accuracy, problem-solving, creativity, collaboration, and reflection. 6. Set checkpoints and deadlines for each phase of the activity (planning, execution, and presentation/submission) to help students manage time and stay organized. 7. Facilitate presentations and sharing where each group presents or displays their output, ensuring that the whole class benefits from the different approaches and perspectives.

	Option 3: Song / Jingle / Rap 1. Form small groups (5–6 members). 2. Compose a short jingle, rap, or song (1–2 minutes) that explains: • The differences between Business Loans and Consumer Loans. • Key words like <i>purpose, repayment terms, requirements, interest rates</i>. 3. Include at least one line that shows a loan computation (e.g., monthly payment). 4. Perform live or record your performance to present to the class. Option 4: Storyboard / Comic Strip 1. Form small groups (5-6 members) 2. Create a 4–6 panel comic strip or storyboard showing a character applying for a loan. 3. Your comic must include: • The character’s reason for borrowing. • The bank officer explains whether it is a Business or Consumer Loan. • The loan’s requirements, repayment terms, and interest rate. • At least one sample loan computation (mortgage or amortization). 4. Submit your comic strip on short bond paper or digitally.	8. Guide reflection by asking students to connect what they created to real-life financial decision-making, highlighting the value of distinguishing between business and consumer loans.
	C.2. Making Generalization	
	Activity C.2: Wrapping up the Lesson Instructions: 1. Look at the word “Loan” written at the center of the board.	Activity C.2: Wrapping up the Lesson This word web activity helps learners consolidate their understanding of loans by generating and organizing key vocabulary related to borrowing. Students collaborate to

	2. Think of words related to loans based on what we learned today (e.g., bank, business, interest, repayment, mortgage, consumer). 3. When called, share your word aloud so it can be added to our word web. 4. Work together as a class to group the words under Business Loans or Consumer Loans. 5. Reflect: How do these words summarize what we have learned about loans?	connect terms to either business loans or consumer loans, reinforcing the differences and common features of each type. The final reflection encourages learners to summarize and articulate the main concepts learned in the lesson. Strategy 1: Brainstorming 1. Write “Loan” in the center of the board. 2. Give students 1–2 minutes to quickly list down words they remember about loans. 3. Call volunteers to share and write their ideas on the board. 4. Group the ideas into categories and connect them to form the web. 5. Highlight the most important keywords that summarize the lesson. Strategy 2: Interactive Discussion 1. Post “Loan” at the center. 2. Ask guiding questions like: • “What words are related to loans based on what we have learned today?” • “Which ones apply to Business Loans? Which applies to Consumer Loans?” 3. Use student responses as a springboard to restate the generalization.
C.3. Evaluating Learning		
	Activity C.3: Assessing Learning Outcomes Instruction: Perform the indicated task. Write your answer on a separate sheet of paper. A. Loan Identification. Write B if the statement describes a <i>Business Loan</i> and C if it describes a <i>Consumer Loan</i>. 1. Borrowed mainly to start, maintain, or expand a business.	Activity C.3: Assessing Learning Outcomes Answer Key A. Loan Identification 1. B 2. C 3. B 4. C



Across

1. A _____ is money lent specifically for business purposes, such as starting or expanding a business.
2. The _____ of loan refers to the length of time given to fully pay a loan.
3. The _____ is the method of paying a loan in equal installments covering both principal and interest.
4. The _____ for consumer loans are often fixed, requiring monthly installments over a set period.

C. Problem Solving

$$1. F = (1 + j)^n = 200,000(1 + 0.08)^3 = P251,942.40$$

$$2. P3,040,000$$

$$3. P1,275,000$$

$$4. B_3 = R \left[\frac{1 - (1 + j)^{-(n-k)}}{j} \right] = 26,379.75 \left[\frac{1 - (1.01)^{-(5-3)}}{0.1} \right]$$

$$= P45,783.04$$

$$5. R = \frac{P}{\frac{1 - (1 + j)^{-n}}{j}} = \frac{30,000}{\frac{1 - (1 + 0.0075)^{-12}}{0.0075}} = P2,623.54$$

5. A _____ is a loan secured by collateral that must be repaid according to agreed terms.
6. Business loans often require documents such as business plans, collateral, and credit history; these are called _____.

Down

1. A _____ is money lent to an individual for personal or family use, such as buying a house or car.
2. The _____ balance is the amount of debt that still remains unpaid at a specific time.
3. The _____ of a business loan is usually to fund capital, equipment, or raw materials.
4. Consumer loans usually have higher _____ than business loans because they are often unsecured.

C. **Problem Solving.** Solve the following problems correctly

1. A loan of P200,000 is to be repaid in full 3 years. If the interest rate is 8% per annum. How much will be paid after 3 years?
2. For a purchase of a house and lot worth P3,800,000 the bank requires 20% down payment, find the mortgaged amount.
3. A car dealer offers a 15% down payment for the purchase of the car. How Much is the mortgage amount if the cash value of the car is P1,500,000?
4. Mr. Ramos is considering paying his outstanding balance after 3 years of payment. The original amount

	of the loan is P100,000 payable annually in 5 years. If the interest rate is 10% per annum and the regular payment is P26,379.75 annually, how much is the outstanding balance after the 3rd payment? 5. A Consumer loan worth P30,000 is to be repaid in 12 months at 9% convertible monthly: How much is the monthly payment?	
C.4. Additional Activities		
	Activity C.4: Extending and Reinforcing Learning Remediation Solve the following problems. Show complete solution. 1. If a condominium is purchased for P5,700,000 and the bank requires 30% down payment, how much is the mortgage amount? 2. A family obtained a P1,000,000 mortgage. If the monthly payment is P38,0000 for four years, how much is the total interest paid? 3. Mr. Canlapan obtained a 10-year mortgage for P4,200,000. If his monthly payment is P39,500, how much is the total interest? 4. A business loan worth P750,000 is to be repaid in quarterly installment in 2 years. How much is the quarterly payment if money is worth 8% converted quarterly? Enhancement Solve the following problems. Show complete solution 1. Ms. Saya got a business loan worth P900,000. She promised to pay the loan semi-annually in 3 years. The semi-annual payment is P183,026.37 if the money is worth 12% converted semi-annually, how much is the outstanding balance after the first payment?	Activity C.4: Extending and Reinforcing Learning This additional exercise may be given as an assignment. Remediation should be provided to learners who have not yet achieved mastery of the lesson, while enrichment activities should be offered to those who are ready for further learning. Answer Key: Remediation 1. P3,990,000 2. P824,000 3. P540,000 4. $R = \frac{P}{\frac{1-(1+j)^{-n}}{j}} = \frac{750,000}{\frac{1-(1+0.02)^{-8}}{0.02}} = P102,382.35$ Enhancement 1. $B_1 = R \left[\frac{1-(1+j)^{-5}}{j} \right] = 183,026.37 \left[\frac{1-(1.06)^{-5}}{0.06} \right] = 770,973.65$ 2. - a.) $R = \frac{P}{\frac{1-(1+j)^{-n}}{j}} = \frac{500,000}{\frac{1-(1+0.01)^{-36}}{0.01}} = P16,607.15$ - b.) $B_2 = R \left[\frac{1-(1+j)^{-34}}{j} \right] = P16,607.15 \left[\frac{1-(1.06)^{-5}}{0.06} \right] = 770,973.65$ - c.) $I_3 = i(P_2) = (0.01)(476,669.63) = P4,766.00$ (for interest) $PR_{51} = R - I_{50} = 16,607.15 - 4,766.7 = P4,766.7 = P11,840.45$ (for principal)

	2. A newly married couple decided to buy a brand new car. The net amount of the loan is P500,000. They plan to amortize the loan in monthly installments for 3 years. If the money is worth 12% convertible monthly, a.) how much is the monthly installment? b.) How much is the outstanding balance after the second payment? c.) How much of the third payment goes to pay the interest and the principal?	
VI. SUMMATIVE ASSESSMENTS	Please refer to the attachments for the summative assessment and performance task.	
VII. REFLECTION	Reflection is most effective when done regularly. Make it a habit to reflect after each lesson or weekly to track your progress and growth over time. The reflection questions are designed to help you thoughtfully evaluate your teaching, identify strengths, recognize challenges, and plan improvements to enhance student learning. The questions provided below may help in reflecting to your teaching performance. 1. What aspects of your teaching went particularly well? • Which part of the lesson engaged students the most? • What teaching strategies or methods contributed to successful learning? • How did students respond to your instruction (e.g., participation, performance, behavior)? • What evidence shows that learning objectives were achieved? • What personal teaching strengths were evident in this session? 2. What challenges did you encounter, and how did you address them? • What difficulties arose during lesson delivery or classroom management? • Were there moments when students struggled to understand a concept? Why? • How did you adjust your strategies or pacing to respond to these challenges? • What support or resources would have helped you handle the situation more effectively? • What will you do differently next time to overcome similar obstacles? 3 What will you do differently next time to improve student learning? • Which teaching strategies or approaches would you modify or try in the future? • How will you better address diverse learning needs in your classroom? • Are there new resources, activities, or technologies you plan to incorporate? • What steps can you take to enhance student engagement or participation?	

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|--|--|
| | <ul style="list-style-type: none">• How will you assess whether these changes improve learning outcomes? |
|--|--|

SUMMATIVE TEST

Directions: Choose the letter of the correct answer. Write only the letter on your answer sheet.

1. What do we call the original amount of money invested or borrowed?
A. Future Value B. Interest C. Principal D. Term
2. What is the total amount after adding the interest to the principal?
A. Compound Interest C. Principal
B. Maturity Value D. Simple Interest
3. Which formula is used in compound interest?
A. $I = Prt$ C. $F = P \left(1 + \frac{j}{m}\right)^{tm}$
B. $F = P(1 + rt)$ D. $I = \frac{F}{p}$
4. If an investment is compounded quarterly, how many times per year is the interest added?
A. 2 B. 3 C. 4 D. 12
5. Which of the following describes simple interest?
A. Interest is computed only on the original principal.
B. Interest is computed on principal and accumulated interest.
C. Interest decreases every year.
D. Interest is given only after maturity.
6. Teacher Ana invests in Pag-IBIG MP2. This is an example of:
A. Loan B. Savings and Investment C. Expense D. Debt
7. An OFW deposits ₱100,000 in a bank. After 5 years, the money grew due to interest. The increase in money is called:
A. Principal B. Future Value C. Interest D. Term
8. If interest is compounded monthly, what is the value of m in the formula?
A. 4 B. 6 C. 12 D. 24
9. Which of the following investments usually gives a higher return because of compounding?
A. Money under the bed C. Bank time deposit
B. Cooperative savings D. Compound interest investment
10. What happens to the maturity value as the compounding frequency increases (monthly vs annually)?
A. It decreases. C. It increases.
B. It stays the same. D. It disappears.
11. Which of the following refers to the fixed sum of money paid to someone at regular intervals and subject to a fixed compound interest?
A. annuity C. simple interest
B. compound interest.. D. annuity certain
12. If the interest conversion or compounding period is unequal or not the same as the payment interval, which type of annuity will be used?
A. annuity certain C. general annuity
B. annuity uncertain D. simple annuity

13. What type of annuity is represented if the payment is made at the end of each month for money borrowed that charges 0.15% interest compounded quarterly?
A. annuity due B. general C. ordinary D. simple
14. For a Simple Annuity with an annual nominal interest rate of 6% compounded quarterly, what is the periodic interest rate i used in the future value formula?
A. 0.0025 B. 0.005 C. 0.015 D. 0.06
15. An investor deposits ₱5,000 at the end of each year for 4 years into an account earning 5% interest, compounded annually. What is the total Future Value of the annuity? (Round to the nearest centavo.)
A. P20,000 B. P21,000 C. P21,550.63 D. P22,050.00
16. Which of the following is referred to as the financial contract with an insurance company that allows you to accumulate funds and defer receiving income payment until a future date, typically in retirement?
A. Annuity due C. Ordinary annuity
B. Deferred annuity D. Simple annuity
17. What are the two distinct periods in the deferred annuity contract?
A. The Initial Phase and the Final Phase
B. The Deposit Phase and the Drawing Phase
C. The Premiums Phase and the Payment Phase
D. The Accumulation Phase and the Annuitization Phase
18. Which of the following is the primary benefit of the "Accumulation Phase"?
A. Tax-deferred growth on all earnings and interest.
B. Guaranteed principal protection from market loss.
C. The exclusion of premium payments from taxable income.
D. Income payments are delayed until the annuitization phase.⁴
19. A client makes a single premium payment of P150,000 into a deferred annuity. If the fund earns an average annual interest rate of 7% compounded annually, what will be the approximate fund value at the end of the 12-year deferral period?
A. P255,000 B. P276,000 C. P337,848 D. P360,980
20. A retirement goal requires an annuity fund value of P1,000,000 after 20 years. If the single premium deferred annuity is projected to earn a fixed annual rate of 6%, what single lump must be invested today?
A. P311,805 B. P371,130 C. P400,000 D. P500,000
21. Which best defines a business loan?
A. Money lent with no interest involved
B. Money lent only to government offices
C. Money lent for personal or family purposes
D. Money lent specifically to fund or expand a business

22. Which of the following is an example of a consumer loan?
- A. Borrowing to buy a family car
 - B. Borrowing to buy factory equipment
 - C. Borrowing to open a new supermarket
 - D. Borrowing for business capital expansion
23. Which statement best differentiates business loans from consumer loans?
- A. Business loans are usually for households; consumer loans are for companies.
 - B. Business loans are for profit-generating purposes; consumer loans are for personal use.
 - C. Consumer loans always have lower interest rates than business loans.
 - D. There is no difference between the two.
24. What is meant by the term of a loan?
- A. The total interest charged
 - B. The method of repayment
 - C. The time given to pay the entire loan
 - D. The collateral used in a mortgage
25. Which of the following correctly describes a mortgage?
- A. A short-term loan with no collateral
 - B. A loan secured by collateral that must be repaid under specified terms
 - C. A loan for government projects only
 - D. An amount borrowed with zero interest
26. Mr. Garcia borrowed ₱1,000,000 at 7% interest to be repaid after one year. How much must he pay in total?
- A. ₱1,050,000 B. ₱1,060,000 C. ₱1,070,000 D. ₱1,100,000
27. A person borrowed ₱1,200,000 for a car through a 5-year mortgage with monthly payments of ₱31,000. How much is the total interest paid?
- A. ₱600,000 B. ₱660,000 C. ₱700,000 D. ₱720,000
28. If a house is worth ₱3,000,000 and the bank requires 20% down payment, what is the mortgage amount?
- A. ₱2,200,000 B. ₱2,300,000 C. ₱2,400,000 D. ₱2,500,000
29. A ₱80,000 consumer loan is to be repaid in 12 months at 8% interest, compounded monthly. What is the approximate monthly payment?
- A. ₱6,500 B. ₱6,959 C. ₱7,200 D. ₱7,500
30. Mr. Tan's 5-year loan requires monthly payments of ₱11,122.22 at 12% interest compounded monthly. What is the outstanding balance after 12 months?
- A. ₱400,000.00 B. ₱415,000.50 C. ₱422,354.73 D. ₱430,000.00

Key Answers

A. Multiple choice.

1. C
2. B
3. C
4. C
5. A
6. B
7. C
8. C
9. D
10. C
11. A
12. C
13. B.
14. C
15. C
16. B
17. D
18. A
19. C
20. A
21. D
22. A
23. B
24. C
25. B
26. C
27. B
28. C
29. B
30. C

Unit 10 - Performance Task:

Objective: By the end of this lesson, learners are able to create a financial proposal using comparisons of different investment plans to recommend and explain the best financial strategy.

Title: Building a Financial Proposal Through Smart Investing

Groupings:

- Learners will work in groups of 4–6 members.
- Each member has a specific role:
 1. Researcher (given data handler): Uses the provided investment data (interest rates, terms, etc.) as inputs.
 2. Mathematician/Analyst: Computes future value, present value, and interest based on the given data.
 3. Data Manager: Creates tables, graphs, and charts.
 4. Writer/Editor: Drafts the financial proposal with explanations.
 5. Presenter(s): Lead the delivery of the group presentation.

Task Description:

As a team of financial consultants, you are tasked to design a financial proposal for a given client profile (see sample cases below). You will compare three investment plans using the provided investment data and recommend the best financial strategy.

Sample Investment Data (Provided by Teacher)

(These are only examples; teacher may adjust as needed)

- Time Deposit: 3.5% annual interest, compounded annually, minimum ₱50,000, term 5 years
- Mutual Fund: Average return of 7% annually, compounded quarterly, moderate risk
- Pension Plan: 5% guaranteed annual growth, compounded annually, long-term (10+ years)
- Stocks (optional): 10% average annual growth, high risk, compounded annually

Sample Client Profiles (Choose or Assign)

- Rosa, age 25, wants to start saving for her retirement at age 60.
- A young couple, both age 30, are saving for their child's college education in 15 years.
- Mark, age 40, wants to invest for a business expansion fund in 7 years.

Process & Output:

1. Use the given investment data (no external research needed).
2. Compute maturity values, present values, and growth using simple/compound interest formulas.

3. Compare risks, returns, and suitability for the client.
4. Create a Financial Proposal (written report or digital presentation).
5. Present findings as a group (10–15 minutes).

Rubrics

Criteria	4 (excellent)	3 (Proficient)	2 (Developing)	1 (Beginning)
Mathematical Accuracy (Group)	All computations are correct.	A few minor errors.	Some errors affecting results.	Many errors
Comparative Analysis (Group)	In-depth, insightful, well-organized.	Adequate comparison, some insights.	Limited depth.	Incomplete or unclear.
Recommendation (Group)	Clear, realistic, fully supported.	Mostly supported, somewhat realistic.	Weak justification	No clear recommendation.
Collaboration & Role Fulfillment (Individual)	Fully contributed, role well-executed.	Contributed with minor lapses.	Uneven contribution.	Minimal or no contribution.
Presentation (Group)	Clear, creative, professional	Organized, mostly clear.	Somewhat unclear.	Disorganized, hard to follow.

